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Learning of salient prey traits explains Batesian mimicry evolution

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Batesian mimicry evolution involves an initial major mutation that produces a rough resemblance to the model, followed by smaller improving changes. To examine the learning psychology of this process, we applied established ideas about mimicry in *Papilio polyxenes asterius* of the model *Battus philenor*. We performed experiments with wild birds as predators and butterfly wings as semiartificial prey. Wings of hybrids of *P. p. asterius* and *Papilio machaon* were used to approximate the first mutant, with melanism as the hypothesized first mimetic trait. Based on previous results about learning psychology and imperfect mimicry, we predicted that: melanism should have high salience (i.e., being noticeable and prominent), meaning that predators readily discriminate a melanistic mutant from appearances similar to *P. machaon*; the difference between the first mutant and the model should have intermediate salience to allow further improvement of mimicry; and the final difference in appearance between *P. p. asterius* and *B. philenor* should have very low salience, causing improvement to level off. Our results supported both the traditional hypothesis and all our predictions about relative salience. We conclude that there is good agreement between long-held ideas about how Batesian mimicry evolves and recent insights from learning psychology about the role of salience in mimicry evolution.

KEY WORDS: **Discrimination learning, generalization, mimicry evolution, salience, two-step hypothesis.**

In Batesian mimicry, a harmless prey eludes predators by being similar to an unpalatable model that the predators have learned to avoid (Bates 1862). A traditional idea about the evolution of Batesian mimicry is the so-called two-step hypothesis (Nicholson 1927; Turner 1977; Joron 2003; Ruxton et al. 2004), according to which a major mutation first produces a rough resemblance to the model, sufficient for predators to generalize their attack avoidance of the model to the new mimic, followed by fine tuning of the mimic's appearance toward more accurate mimicry. The process has been investigated using genetic analysis of interspecific crosses (Clarke and Sheppard 1955, 1959, 1963), experiments with artificial prey (Mappes and Alatalo 1997), and individual based evolutionary modeling (Franks and Sherratt 2007; Balogh et al. 2010; Gamberale-Stille et al. 2012). There are also a number of experiments with artificial prey supporting the general idea that an imperfect resemblance to a model can reduce predator attacks (e.g., Schmidt 1958, 1960; Morrell and Turner 1970; Ford 1971; Shideler 1973; Aronsson and Gamberale-Stille 2008, 2012). Here, we examine a scenario proposed by Clarke and Sheppard (1959) for mimicry evolution in the eastern black swallowtail, *Papilio polyxenes asterius*, by performing experiments with wild blue tits and butterfly wings.

Our approach is to take established ideas of the evolutionary genetics of mimicry in *P. p. asterius* (Clarke and Sheppard 1955, 1959; Sperling 1987) at face value and examine if the proposed scenario is in agreement with recent ideas about predator psychology. We specifically test if the proposed evolutionary sequence is supported by the hypothesis developed by Kazemi et al. (2014, 2015), entailing that prey traits have different salience to predators. Traits that are perceived as highly salient by the predators will be used primarily in the discrimination and generalization between prey, whereas traits with lower salience will be overshadowed and have low, or no, influence on the prey assessment. According to this hypothesis, mimicry of a single trait that is perceived as highly salient in the given context should elicit a

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considerable amount of predator avoidance due to generalization of the model appearance.

The psychological concept of salience describes aspects of stimuli that are more intense or striking and therefore stand out from the background or in comparison to other stimuli, and attract most of the attention of the observer (e.g., the predator). A learned association will quickly develop between a salient stimulus (e.g., a prey trait) and the outcome it predicts (e.g., reward or punishment). However, salience is not only a consequence of the intensity or background contrast of a stimulus, it can also depend on how discriminable it is from the other stimuli (see Frieman and Reilly 2015 for an introduction to the concept of salience in learning psychology). This means that a stimulus that is not particularly striking or noticeable in itself can be perceived as salient in comparison to the other stimuli due to its discriminability. A prey trait that is easily discriminable between profitable and unprofitable prey will thus have high salience to the predators, given that it is a reliable predictor of the outcome. It should be noted that the salience of a stimulus is not fixed, it is influenced by the context in which it is present and the experience of the learning individual (Mackintosh 1975; Pearce and Hall 1980; Hall and Rodriguez 2010; Kazemi et al. 2015).

Our main contribution here is that we experimentally test a proposed scenario of two-step Batesian mimicry evolution and we specifically show that the concept of salience, derived from learning psychology, can explain three major issues: the initial appearance of a mimic, its further refinement, and the point at which mimicry evolution levels off. Our work is the first direct experimental examination of a scenario of mimicry evolution.

From crosses and backcrosses of *P. p. asterius* and *Papilio machaon*, Clarke and Sheppard (1955) discovered that the black ground-color of *P. p. asterius* is determined by an autosomal locus, and they argued that the melanic allele at this locus was the mutation in *P. p. asterius* that initiated the evolution of mimicry of the unpalatable model *Battus philenor* (Clarke and Sheppard 1959). It is estimated that *P. polyxenes* diverged from its *P. machaon* like sister species *Papilio zelicaon* in Pliocene time (Sperling 1987; Sperling and Harrison 1994; Zakharov et al. 2004; see also Dupuis and Sperling 2015 on the evolution of the North American *P. machaon* group swallowtails), but *P. p. asterius* may have emerged as a mimic of *B. philenor* at a considerably later time, perhaps during the late Wisconsinan glaciation when the species occurred in refugia in southeastern North America (Sperling 1987; Fordyce and Nice 2003). Their current mimic–model relationship is supported by experiments (Brower 1958; Jeffords et al. 1979; Codella and Lederhouse 1989) and field surveys (Brower and Brower 1962).

The ventral wing patterns of male and female *P. p. asterius* are similar to each other and to those of *B. philenor* (Codella and Lederhouse 1989; Hazel 1990; Herrell and Hazel 1995), whereas the dorsal wing surfaces of males are less similar to the model than the female wings. Both *B. philenor* (Pegram et al. 2012) and *P. p. asterius* (Codella and Lederhouse 1989, 1990) expose the ventral wing surfaces during roosting, when protection from predator attacks is particularly important, and this could explain a higher mimetic perfection of these surfaces (Codella and Lederhouse 1989). Based on this, we used ventral hindwings as prey stimuli in our experiments. We let wild-caught blue tits learn to discriminate between rewarded *P. machaon* and unrewarded *B. philenor* prey stimuli, after which we examined how the birds generalized their avoidance of *B. philenor* to other stimuli, including wings of *P. p. asterius* and of hybrids between *P. p. asterius* and *P. machaon*. The latter prey stimulus was intended to represent the first mimetic mutant. To investigate the role of black coloration, we also examined the tendency of the birds to avoid black-painted *P. machaon* wings. In the experiment, *P. machaon* was used as a representative of the general appearance of palatable yellow *Papilio* butterflies, of which *P. zelicaon* is an example.

We put the results of this experiment into the framework of the learning-based explanation of imperfect mimicry developed by Kazemi et al. (2014, 2015), by estimating the salience of the different wing coloration traits. In learning psychology, the salience of a conditioned stimulus is directly connected to the associability of that stimulus in such a way that the speed by which a stimulus is learned is an indication of its salience (Pavlov 1927; Mackintosh 1975; Pearce and Hall 1980; Frieman and Reilly 2015). According to Kazemi et al. (2014, 2015), the first mutation that produces a successful but imperfect mimic should occur in a trait that is perceived as highly salient by the predators in the given context, leading to the prediction that the black ground-color in the current scenario should have high salience, that is, elicit fast discrimination learning. Additionally, the differences between the first mutant and the model should have some degree of salience, leaving scope for further fine-tuning of the similarity to the model, which we predict would be the case with the hybrid prey in our study. Finally, when the remaining differences between the model and the mimic are in traits with very low salience, selection for more accurate mimicry should level off, thus, the differences between *P. p. asterius* and *B. philenor* wings are predicted to have very low salience. We tested these predictions experimentally, again using wild blue tits and butterfly wings. We also measured the amounts of black, orange, and yellow-white coloration in the wing patterns of *B. philenor*, *P. a. asterius*, *P. machaon*, and hybrids to provide quantitative estimates of the trait changes involved in the evolutionary process. Blue color measurements and comparisons were made between the Papilio butterflies. The blue color of the *B. philenor* model is iridescent and not comparable.

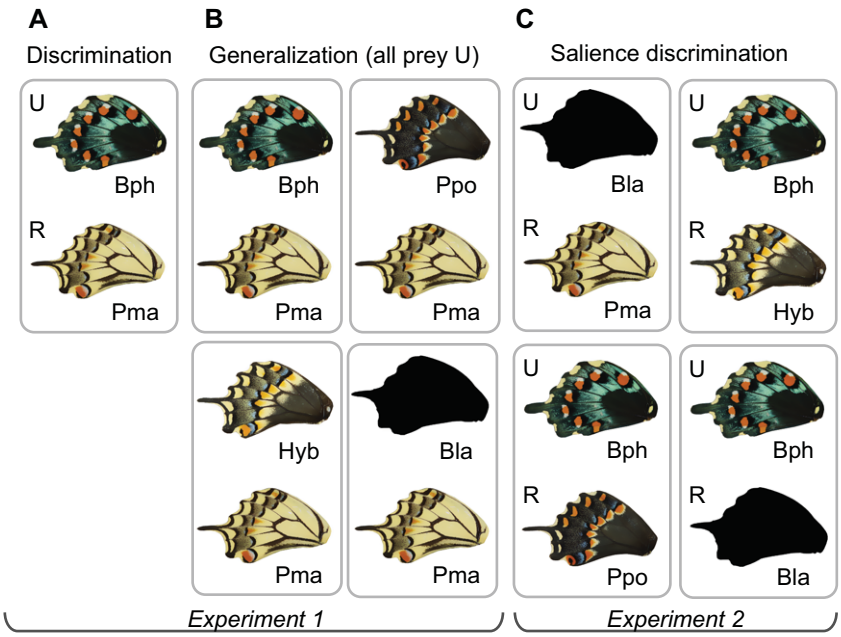

Figure 1. Overview of the prey stimuli and setups used in the experiments. One male wing example per respective stimulus type is shown (see Fig. S3 for additional examples of hindwings). The abbreviations U and R denote if the prey were unrewarded or rewarded in the setup. Bph = *Battus philenor*; Ppo = *Papilio polyxenes asterius;* Pma = *Papilio machaon*; Hyb = hybrid; Bla = black-painted. (A) The treatment in the discrimination learning of Experiment 1, (B) the four treatments in the generalization test in Experiment 1, and (C) the four treatments in the salience test of Experiment 2.

Methods

PREDATORS, PREY, AND ARENA

We used wild blue tits (*Cyanistes caeruleus*) as predators, which we caught with trapping cages and housed indoor during trials, and subsequently ringed and released. All birds ($N = 73$) were included regardless of sex or age. They performed one to two trials per day and were food deprived between 30 and 60 min before each trial to enhance foraging motivation. The experiments were carried out in accordance with regulations on handling of wild birds, with permission from Linköpings djurförsöksetiska nämnd (Dnr 91-12).

The prey stimuli consisted of dried butterfly hindwings of male and female *B. philenor*, *P. p. asterius*, *P. machaon*, F1 offspring hybrids of *P. p. asterius* and *P. machaon*, and *P. machaon* wings painted with black water-soluble paint (Fig. 1). The *B. philenor* specimens were reared in the laboratory by Ron Rutowski (Arizona State University, Tempe, Arizona), and all the *Papilio* butterflies were reared in our laboratory at Stockholm University. The ventral side of the wings was exposed to the birds during the experiments, and the dorsal side was covered with paper that had the same gray color as the background boards on which they were presented (Fig. S1). The wings were also laminated with flexible clear plastic to prevent damage during trials. The reflectance spectra of the black, orange, yellow-white, and blue colors were measured on a *B. philenor* wing, with an Ocean Optics (Dunedin, FL, USA) USB2000 spectrophotometer and a PX-2 pulsed xenon light source, before and after plastic cover was applied, to examine possible effects. The measurements showed that the distribution of wavelengths was quite similar with and without plastic cover (Fig. S5). We used Philips TL-D90 Graphica Pro 36W/950 fluorescent tube lamps in the experimental rooms to simulate daylight and the irradiance spectrum was measured using an Avaspec-2048-USB2-UA spectrometer with a cosine corrector. The irradiance values, together with receptor sensitivity values of blue tits (Fig. S4), were used to estimate the color reflectance values for the wing colors (Fig. S5).

The trials were carried out in small experimental rooms equipped with a bird stand and one-way mirrors for observation and filming. The birds were released into the room through a hatch and were free to fly around and forage from experimental boards (gray-painted MDF-boards) that were situated on the floor.

EXPERIMENTAL PROCEDURES

The general approach was similar to that in Kazemi et al. (2014). During trials, the prey stimuli were presented on top of small wells in the experimental boards (Fig. S1). The wells beneath the profitable prey were prepared with a piece of mealworm (*Tenebrio molitor)* as a reward, and the wells beneath the unprofitable prey were left empty as negative feedback (see, e.g., Alcock 1970). The background boards had 30 wells in total, of which half was covered with profitable prey and half was covered with unprofitable

prey during trials. The prey items were shuffled before each trial to avoid spatial learning. The birds could only access the well and the reward by lifting the prey wing (regarded as an attack). This behavior was learned during a pre-experimental phase including training with white square paper items covering wells to different degrees, both in the holding cages and in the experimental rooms.

The first part of Experiment 1 consisted of a discrimination learning task in which the birds were expected to learn to avoid the unprofitable model prey during five trials. They were presented with 15 *B. philenor* (Bph) wings that were unrewarded model prey, and 15 *P. machaon* (Pma) wings that were rewarded profitable prey (Fig. 1A). The second part of the experiment consisted of a one-trial generalization test in which we replaced the unrewarded model prey with one of following mimic wings (one treatment per bird): *P.p. asterius* (Ppo), hybrid of *P. p. asterius* and *P. machaon* (Hyb), and black-painted *P. machaon* (Bla). One group was tested with the model prey as a control (Fig. 1B). All prey were unrewarded in the generalization test, and the sample sizes for the treatments were $N = 10, 9, 9, 9$, respectively.

Experiment 2 was a salience test consisting of a discrimination learning task, much similar to Experiment 1, with new birds that learned to discriminate between rewarded and unrewarded prey during six trials. We used four discrimination treatments: Bph-Ppo, Bph-Hyb, Bph-Bla, and Bla-Pma ($N = 9$ for each treatment). Each bird was assigned to one discrimination treatment. In the first three treatments, the model (Bph) was unrewarded, and in the fourth treatment the black-painted wings (Bla) were unrewarded (Fig. 1C). The rate by which the birds learned to discriminate between the rewarded and unrewarded prey was used as an indication of how salient they perceived the differences between their appearances.

WING COLORATION MEASUREMENTS

For each wing type (Fig. S3), we chose five female and five male individuals that were photographed on both the dorsal and ventral side using a Nikon D 5100 camera with an AF-S Micro Nikkor 60 mm lens and a Sigma EM 140DG flash. The photos were processed and analyzed using the software GIMP 2.8.14 (2014). We extracted the wing area from the background and isolated the total number of colored pixels. By using the "select by color" tool (threshold 75 of 255, composite), we separated the differently colored areas of the wing into the categories black (i.e., melanized), orange, yellow-white, and blue, and measured the number of pixels they covered. This was done by selecting a central location of the specific color, and due to the set threshold, all area covered by the color was selected, including edge pixels. Very similar coverage would have been reached by measuring on the edge of color areas, as long as the approximately same hue was selected. The proportion of the wing area that each color covered was calculated based on the pixel data and used in the analyses (Figs. S3 and S6; Table S3). The blue coloration was measured and analyzed for the *Papilio* butterflies, but not for the *B. philenor* model. The iridescent blue of the model can vary between laboratory-reared and wild specimens (Rutowski et al. 2010), and be strongly affected by the photographic method in comparison to the *Papilio* butterflies.

STATISTICAL METHODS

We analyzed the data using the statistical software R version 3.3.1 (R Core Team 2016). Data from discrimination learning trials, both from Experiments 1 and 2, were analyzed by fitting a generalized linear mixed-effects model using the glmer function in the lme4 package (Bates et al. 2015). The number of attacks on rewarded prey, out of the $n = 10$ first attacks in a trial, was used as a binomially distributed response variable. For the analysis of the discrimination learning task in Experiment 1, trial was used as fixed effect, and for the analysis of the data from Experiment 2, trial, treatment, and their interaction were used as fixed effects. In both cases, the individual bird was used as random effect. The generalization test consisted of a single trial, so for this analysis we used the glm function to fit a generalized linear model with the number of attacks on the learned profitable prey type, out of the $n = 10$ first attacks, as response variable, and treatment as fixed effect. For post hoc testing, we used the lsmeans package (Lenth 2016), with Tukey adjustment of P values. Data and R files used for the statistical analyses are deposited in the Dryad Repository.

Results

The aim of Experiment 1 was to investigate how the birds generalized between the unprofitable model prey and different mimics. The birds were first subjected to a discrimination task in which they learned to discriminate and avoid the unrewarded model *B. philenor* and attack the rewarded *P. machaon* (Fig. 1A). Over the course of five trials, with 15 of each prey type in each trial, the birds learned to avoid the model prey, as shown by the positive mid-trial intercept and slope of a mixed-model logistic regression (logit-scale intercept: 1.70 ± 0.12, $P < 0.0001$; slope: 0.77 ± 0.07, $P < 0.0001$; Fig. S1). We then tested how the birds generalized between the avoided model and different mimics (Fig. 1B), including *B. philenor* ("perfect mimic" as control), *P. p. asterius*, hybrids of *P. p. asterius* and *P. machaon* ("first mutant"), and black-painted *P. machaon*. In the setup, unrewarded mimics of a given type were presented together with an equal number (15) of unrewarded *P. machaon*. Fitted generalized linear models with the number of attacks on nonmimic prey (i.e., *P. machaon*) out of the first n attacks as response variable and mimic type as factor showed, both for $n = 7$ and 10, that the hybrids were statistically significantly less avoided than the control (*B. philenor*), whereas

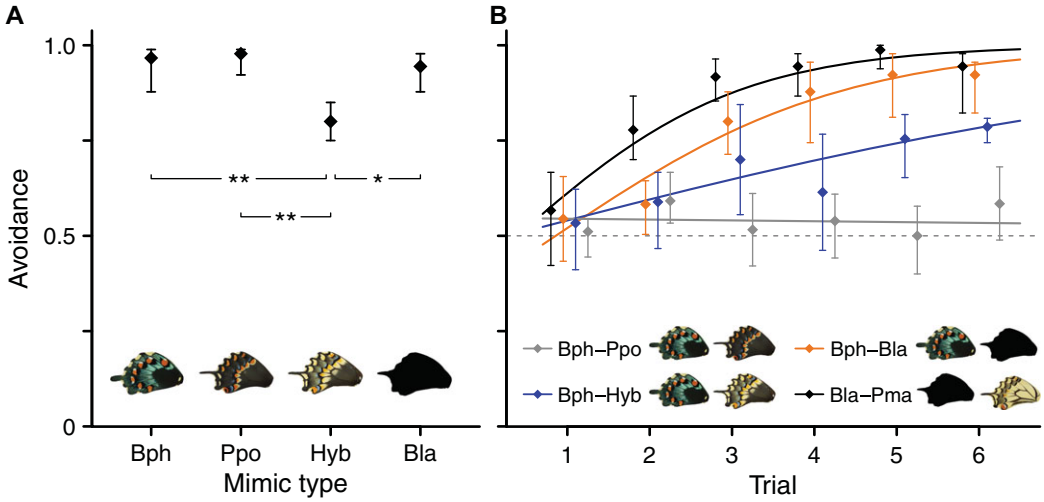

Figure 2. Avoidance of prey stimuli in the generalization and salience tests. The avoidance is shown as the proportion of attacks directed at the rewarded prey, that is, away from unrewarded prey, and is given as the mean proportion with 95% bootstrap confidence intervals. Panel (A) shows the avoidance of the model (*Battus philenor*) and the three mimic wing types (*Papilio polyxenes asterius*, hybrid, and black) in the one-trial generalization test ($N = 37$), with statistically significant differences marked with stars ($^{*}P \leq 0.05$, $^{**}P \leq 0.01$; see Tables S1 and S2 for statistical analyses). Panel (B) shows the progress of the four discrimination treatments in the salience test ($N = 36$) as a function of trial number, together with fitted logistic regression curves, that is, learning curves. The model (Bph) is unrewarded in all treatments, and Bla is unrewarded in the Bla-Pma treatment. Based on pairwise post hoc comparisons of midtrial intercepts and slopes, each curve was statistically significantly different from the others (Table 1).

P. p. asterius and black-painted wings did not differ from the control in how strongly they were avoided (Table S1). The avoidance of mimics is illustrated (for $n = 10$) in Figure 2A, together with statistical significances from a Tukey post hoc test (Table S2).

To estimate the salience of differences in prey appearance as perceived by the birds, we performed Experiment 2, in which new birds were given discrimination tasks similar to the one in Experiment 1, with six trials including 15 each of unrewarded and rewarded prey. There were four treatment groups (Fig. 1C): black-painted wings versus *P. machaon*, *B. philenor* versus hybrid, *B. philenor* versus *P. p. asterius*, and *B. philenor* versus black-painted wings. The first three treatments test the predictions that the black color difference should have high salience, that the trait differences between the model and the "first mutant" (with partial melanism) should have intermediate salience, and that the trait differences between the model and the current mimic should have very low salience. The final treatment investigated whether the avoided black-painted wings really were good mimics of the model. The results, based on the first 10 attacks in a trial, are shown in Figure 2B and support our three predictions. Post hoc statistical tests showed that each of the learning curves in Figure 2B differs from the others (Table 1). In addition, the birds rapidly learned to discriminate between *B. philenor* and black-painted wings (Fig. 2B). This shows that even though black wings were strongly avoided following *B. philenor*–*P. machaon* discrimination learning (Fig. 2A), they were not good mimics of *B. philenor*.

The ventral hindwing melanism of females of *P. machaon*, *P. p. asterius*, and hybrids is illustrated in Figure 3. Hybrids are heterozygotes with one melanic and one "yellow" allele and are intermediate but closer to *P. p. asterius* than to *P. machaon* in melanism (Figs. 3d, S3, and S6; Table S3). Fitting a linear model to the melanized proportion of the ventral hindwings showed that *B. philenor* males were more melanized than females (Fig. S6a; $F_1 = 27.5$, $P < 0.0009$). The *Papilio* ventral hindwings, that is, *P. p. asterius*, hybrids, and *P. machaon*, differed significantly in melanism, but there were no statistically significant sex differences (Table S3). For the proportion orange on ventral hindwings, females of *B. philenor* had more orange than males (Fig. S6c; $F_1 = 6.0$, $P = 0.04$), and for *Papilio*, hybrids were approximately halfway between *P. p. asterius* and *P. machaon* (Fig. S6c; Table S3), and there were no statistically significant sex differences. For the proportion of blue on ventral hindwings of the *Papilio* butterflies, the sexes were not significantly different but there was a nonsignificant tendency for *P. machaon* to have a lower proportion of blue (Table S3).

Discussion

In the scenario proposed by Clarke and Sheppard (1959, p. 255), *P. polyxenes* evolved mimicry of the model *B. philenor* in a sequence of steps, the first being the appearance of a mutant melanic allele of major effect, producing tolerable mimicry, followed by several smaller changes in wing coloration that increased the

Table 1. Tukey lsmeans post hoc comparisons for the mixed-effect generalized linear model analysis of the salience test in Experiment 2.

Comparison	Intercept	SE	*P*	Slope	SE	*P*
Bph-Hyb–Bph-Ppo	0.561	0.197	**0.022**	0.212	0.077	**0.029**
Bph-Bla–Bph-Ppo	1.351	0.213	**<0.0001**	0.561	0.091	**<0.0001**
Bla-Pma–Bph-Ppo	2.152	0.255	**<0.0001**	0.732	0.115	**<0.0001**
Bph-Bla–Bph-Hyb	0.790	0.217	**0.002**	0.348	0.094	**0.001**
Bla-Pma–Bph-Hyb	1.591	0.257	**<0.0001**	0.519	0.117	**0.0001**
Bla-Pma–Bph-Bla	0.801	0.270	**0.016**	0.171	0.127	0.533

The comparisons are between mid-trial intercepts and slopes of the four discrimination treatments; see Figure 2B for fitted learning curves. Bold values denote statistically significant differences ($P < 0.05$).

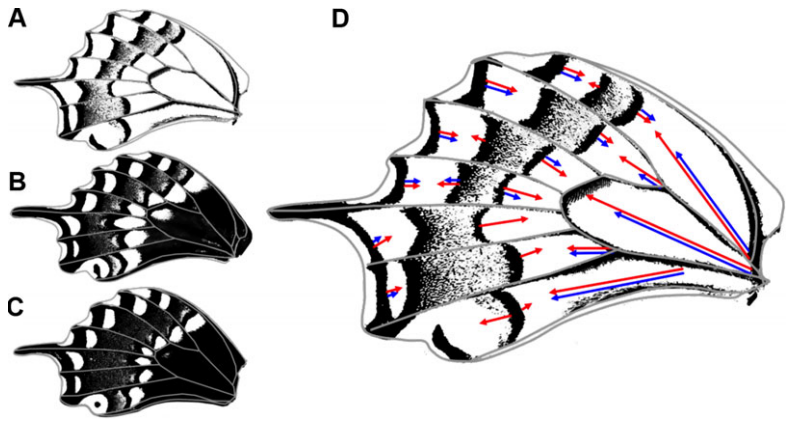

Figure 3. Illustration of the melanized areas of the ventral side of the hindwings of a *Papilio machaon* female (A), a hybrid female (B), and a *Papilio polyxenes asterius* female (C). The arrows in (D) show estimates of evolutionary changes in melanism from a *P. machaon* coloration to that of the hybrid (blue arrows) and to that of *P. p. asterius* (red arrows). The arrows represent increased melanism of interspaces on the wing, expanding from the *P. machaon* melanized wing veins and interspaces.

mimetic perfection. Our experiments support this scenario, in the sense that the hybrid wings, which we assume approximate the appearance of the first mimetic mutant in *P. p. asterius*, achieved substantial but not perfect protection from attacks by the birds that had learnt to discriminate between *B. philenor* and *P. machaon* (Fig. 2A). These birds also strongly avoided attacking black-painted wings in accordance with the idea that the melanism of the hybrids was important for their protection. It should be noted that the birds in our experiments did not have an inherent tendency to avoid black-painted wings, nor any other of our prey stimuli for that matter, as can be seen from the first trial of the salience test (Fig. 2B). The avoidance of black wings in the generalization test (Fig. 2A) was thus a result of the birds learning to discriminate *B. philenor* from *P. machaon* (Fig. 1A).

The learning-based explanation of imperfect mimicry by Kazemi et al. (2014, 2015) appears to work well for this instance of mimicry evolution. The major mutation suggested to initiate mimicry evolution increases wing surface melanism (Fig. 3) and our experiment showed that this trait is of high salience for the birds (Fig. 2B). Furthermore, the "first mutant" (hybrid wings) turned out to be protected only to some extent (Fig. 2A), consistent with the intermediate salience of the hybrid—*B. philenor* difference (Fig. 2B), leaving scope for further perfection of the mimicry. Finally, the birds failed to learn to discriminate *P. p. asterius* from *B. philenor* even after six trials each including 30 prey stimuli, which shows that this difference in ventral hindwing color pattern had very low salience for the birds, even though it may seem relatively clear to human eyes (Fig. S3). Thus, the concept of salience contributes importantly to explain the sequence proposed by Clarke and Sheppard (1959) for the evolution of mimicry in *P. p. asterius*. Several aspects of the evolution of Batesian mimicry, including the role of imperfect mimicry, have been understood for quite some time, as presented in the review by Turner (1977) from 40 years ago. Our experiments complete and explicate the idea by showing that the learning psychological concept of salience provides an explanation for phenomena relating to imperfect mimicry that played an important part for the two-step hypothesis. This includes explaining the original appearance of a mimic, its further refinement, and the point at which mimicry evolution levels off. Also, our work is the first to experimentally examine a specific proposed scenario of mimicry evolution.

There is a tradition in mimicry research to use a general concept of mimic–model similarity, as exemplified by expressions as "rough similarity" or "perfect similarity." Our perspective here, examining the relative salience of traits in discrimination learning, provides a more precise description of the psychological mechanism, beyond the intuitive idea of similarity. For instance, the black-painted wings appeared to be good mimics in the generalization test of Experiment 1 (Fig. 2A) in the sense that they were avoided as much as the model. A likely reason is that the birds had learnt to associate black with unrewarded prey. Nevertheless, in Experiment 2, the birds readily learnt to discriminate black-painted wings from *B. philenor* wings (Fig. 2B), so the black-painted wings were not good mimics in a situation where the birds had extended exposure to both mimic and model. An advantage of using the concept of salience, in comparison with general similarity, is that salience can depend on the context of the particular discrimination.

The *P. p. asterius–P. machaon* hybrid wings used in our experiments approximate the appearance of a melanic "first mutant," but they might have had a somewhat greater proportion orange, and possibly also blue coloration than the actual first mutant (Figs. S3 and S6; Table S3). Our crosses show that the melanic allele is not fully dominant (Figs. 3, S3, and S6; Table S3), although its effect is strong enough that Clarke and Sheppard (1955, 1959) referred to it as a dominant gene. At any rate, if the first melanic individuals were protected enough for the allele to increase in frequency, the consequence would have been increased melanism as homozygotes were formed, increasing the mimetic effect. More crosses, including backcrosses to disentangle genetic effects on melanism, orange and blue coloration, together with additional experiments with bird predators, would be needed to better estimate the relative importance of black, orange, and blue for the mimetic function. Experiments both in the laboratory and the field on the possible importance of orange and blue coloration for aposematism in *B. philenor* have shown that orange and blue both contribute to aposematism (Pegram et al. 2013) or that orange possibly contributes more (Pegram et al. 2015). Considering our experiments, melanism appears to be the dominating component of the *B. philenor* warning signal. If predators have the opportunity to gain more experience with both the model and palatable mimics, as happened in our salience tests, the orange and possibly the blue components may have an effect on the discrimination of prey.

In our experiments, we used lack of reward to indicate unprofitable prey (i.e., empty wells under prey stimuli). From our experience with birds attacking artificial prey, unrewarded versus rewarded prey serve as effective reinforcers in discrimination learning (e.g., Kazemi et al. 2014, 2015), but there is also a tradition to use distasteful items to indicate unprofitable prey (e.g., Schmidt 1958, 1960; Morrell and Turner 1970; Aronsson and Gamberale-Stille 2008, 2012). The choice of negative reinforcement may influence the rate of discrimination learning, but more important for our results here, it could influence how birds generalize. There is experimental support for the idea that birds generalize more widely after discrimination learning with more distasteful models (Duncan and Sheppard 1965; Goodale and Sneddon 1977); however, there are also studies where the effect was relatively weak (Lindström et al. 1997). In a comparative study by Alcock (1970), empty seeds and emetic seeds both elicited rapid and consistent avoidance, even in the absence of rewarded stimuli. Also, the effect of model averseness on prey generalization seems highly correlated with the mimic:model frequencies. Based on signal-detection theory, the effect of wider generalization due to the costliness and frequency of the model is to be expected (Sherratt 2002). Thus, using distasteful rather than unrewarded model prey in our experiments could have, for instance, reduced the attacks on hybrid wings somewhat, but that would not have changed our qualitative conclusions.

Blue tits are sensitive to ultraviolet light and have a Ultraviolet sensitive (UVS) color vision system (Ödeen and Håstad 2003; Endler and Mielke 2005; Hart and Hunt 2007; Lind et al. 2014). Both of the major bird color vision systems, UVS and VS, are likely to be represented among insectivorous birds that may prey on *B. philenor* and its mimics, considering the bird species found to attack *B. philenor* in the field (Pegram et al. 2015). The vision system could influence the sensitivity to the blue coloration. Up to now, it is not known what the effects of predator color vision systems might be on mimicry evolution, although it has been suggested that more longwave-rich colors tend to be the most important for warning coloration (Stevens and Ruxton 2012).

A reason for thinking that mimicry evolved more recently in *P. p. asterius* than at the time of origin of *P. polyxenes* is that not all members of the species are mimetic. For instance, the subspecies *P. p. americus* and *P. p. stabilis*, which occur in South and Central America, show varying degrees of melanism (Warren et al. 2012) and are sometimes similar in appearance to *P. zelicaon* and *P. machaon*. The same holds for the subspecies *P. p. coloro*, which occurs in southwestern North America (although in this case one cannot exclude that the yellow morph genes have introgressed from adjacent *P. zelicaon*). It is then likely that mimicry evolved in *P. polyxenes* from an appearance similar to that of *P. machaon*. Using genomic data to better determine when and where this took place could be a promising avenue of investigation.

The evolutionary genetics of mimicry has advanced greatly in recent years (Reed et al. 2011; Martin et al. 2012; Gallant et al. 2014; Kunte et al. 2014; Timmerman et al. 2014; Nishikawa et al. 2015), with studies on both Müllerian and Batesian mimicry. In the mimicry ring centered on *B. philenor* as a model, *Limenitis arthemis* can be compared with *P. polyxenes*, in that Batesian mimicry of *B. philenor* has evolved relatively recently, around

200,000 years ago, perhaps in glacial refugia (Mullen et al. 2008), forming the mimetic subspecies *L. a. astyanax*. A melanic mutant appears to have been the first step of this mimicry evolution, and the molecular basis of the mutant is well characterized (Gallant et al. 2014), although the genetic basis for subsequent smaller improvements in mimicry is not known. These examples of relatively recently evolved mimicry are prime candidates for investigating basic evolutionary principles such as the two-step hypothesis, ideally using both molecular genetics and studies of predator–prey interactions, as in our experiments here.

AUTHOR CONTRIBUTIONS

All authors took part in the design of the study and the interpretation of results. CW reared and hybridized the *Papilio* butterflies, BK and TW performed the experiments, GGS and BK performed the wing coloration measurements, OL and BK performed the statistical analyses and wrote the manuscript, with modifications provided by the other authors.

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DATA ARCHIVING

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's website:

Figure S1. Experimental arena, view from above.

Figure S2. Discrimination learning of unrewarded model prey (*Battus philenor*) and rewarded prey (*Papilio machaon*), which are shown in Figure 1A.

Figure S3. Examples of ventral and dorsal hindwing color patterns of males and females of *P. machaon* (Pma), hybrids between *Papilio machaon* and *Papilio polyxenes asterius* (Hyb), *P. p. asterius* (Ppo) and *Battus philenor* (Bph).

Figure S4. Estimated spectral curves for the light measured in the experimental box (A) and the receptor sensitivities for blue tit color vision (B).

Figure S5. Color reflectance measurements of a *Battus philenor* hindwing before and after plastic cover.

Figure S6. The proportion of hindwing surfaces that is either melanized (A, D), yellow-white (B, E), or orange colored (C, F), shown for ventral and dorsal hindwing sides of females (red round symbols) and males (blue triangle symbols) of *Papilio machaon* (Pma), hybrids between *P. machaon* and *Papilio polyxenes asterius* (Hyb), *P. p. asterius* (Ppo) and *B. philenor* (Bph), which are displayed in Figure S3.

Table S1. Summary of fixed effects from the generalized linear model analysis of the one-trial generalization test in Experiment 1, based on the first 10 attacks.

Table S2. Tukey lsmeans post hoc comparisons for the generalized linear model analysis of the generalization test in Experiment 1.

Table S3. Summary of fitted linear models of the proportions melanized, orange and blue on ventral hindwings of females and males of *Papilio polyxenes asterius*, hybrids, and *Papilio* machaon (see also Fig. S6).